

Prasanna Jonnadula

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EDUCATION

GEORGE MASON UNIVERSITY M.S. IN COMPUTER SCIENCE

GPA: 3.93 / 4.0 | Jan 2024 – Dec 2025

JAWAHARLAL TECHNOLOGICAL UNIVERSITY

B.TECH IN COMPUTER SCIENCE

GPA: 3.8 / 4.0 | 2018 – 2022

SKILLS

Technical Skills:

C, C++, Python, Java, HTML, CSS, Bash script, PHP, React, Node.js

Databases:

Oracle, PostgreSQL, MySQL, MongoDB, NoSQL, PL/SQL, InfluxDB, SQLite.

General Skills:

Generative AI, LLM, Machine Learning, Data structures and Algorithms.

Tools/Libraries:

Tableau, Power BI, Qlikview, QlikSense, Jenkins, Git, JIRA, Visual Studio, IntelliJ IDEA, Pycharm, EC2, Icinga, Zabbix, Grafana, Kafka, RabbitMQ.

Python Libraries:

NumPy, Pandas, PyCurl, urllib3, argparse, Kafka

Cloud & Tools

AWS (EC2, S3, IAM, RDS, DynamoDB), Docker, Jenkins, Kubernetes, Tableau

Operating Systems

Linux, Unix, Windows, macOS

PROJECTS

Blog Assistant with Dalle 3

- LLM model Provides the customised blogs based on provided keywords and topic.
- Connected to Gemini 1.5 Pro and Open API via keys and queries the data and uses Dalle-3 to retrieve images for blogs.
- Applied generative AI techniques.

Smart Budget Bot

AI-driven system leveraging LLMs to analyze USAspending.gov data and generate insights securely on **Tableau**.

Tech: OpenAI, Python, Streamlit, Tableau

Link: Devpost

Vehicle Rental Management System

- Built a MongoDB + Python system with automated ETL pipelines for rental transactions and customer data.
 - Designed normalized schema and REST APIs enabling real-time analytics dashboards.
- Link:** GitHub

EXPERIENCE

GEORGE MASON UNIVERSITY | DATABASE ARCHITECT | JAN 2025 – PRESENT

- Designed and optimized a **PostgreSQL** data warehouse supporting 600K+ records, improving query performance by **50%** through indexing.
- Built automated ETL pipelines (Pandas, SQLAlchemy), reducing manual work **70%**.
- Developed secure REST APIs (FastAPI) for internal analytics and RBAC.
- Built a full-stack analytics platform with **React, Python, PostgreSQL, Docker**, deployed via **CI/CD** on **AWS EC2** for real-time data access.

COMMUNITY ASSOCIATIONS INSTITUTE | DATA SCIENTIST | JUN 2025 – OCT 2025

- Implemented scalable data processing pipelines using **PySpark** and **Python**, processing 300K+ records, reducing runtime by **40%**.
- Built modular data transformation and re-architected data systems to enable weekly production releases, increasing deployment frequency **4x**.
- Built distributed workflows across spatial datasets ensuring high scalability and Created **30+ Tableau dashboards** for housing data.
- Authored APIs, technical documentation, cutting onboarding time.

ACCENTURE | DATA ENGINEER | AUG 2022 – OCT 2023

- Built **50+ Tableau dashboards** by integrating Oracle MySQL pipelines, delivering real-time business insights for Kaiser Permanente.
- Migrated MySQL to InfluxDB, improving scalability and reducing CPU usage **30%**. Worked in an **Agile SCRUM methodology**.
- Automated report delivery and scheduling using **Tableau Server**, reducing business decision latency by **40%**.
- Tuned ETL scripts/SQL queries, improving dashboard refresh rates.

COGNIZANT | FULL STACK INTERN | JAN 2022 – AUG 2022

- Developed backend services to retrieve and process data from **AWS S3**, supporting real-time analytics across 100+ intersections.
- Migrated legacy DB to AWS, optimizing storage and building admin dashboards. Added **Custom triggers** in Zabbix for better monitoring.
- Created **D3.js** visualizations for geospatial mapping and data analysis.

HONORS & AWARDS

10/2025 **AWS Certified Cloud Practitioner.** link: AWS

02/2025 **Certified Tableau Desktop Specialist.** link: Tableau

10/2024 **1st place** – Salesforce & Fiscal Services Hackathon, GMU. link: Devpost

06/2023 **ACE Award** for outstanding team contribution, Accenture

06/2023 **Microsoft Certified:** Azure Fundamentals (AZ-900). link: Microsoft

10/2021 **2nd rank** – B.Tech CSE.

07/2018 Received **7 times Merit scholarship** from NSF For best rank, APEAMCET.

PUBLICATIONS

LINK: PREDICTION OF POPULATION DENSITY & POVERTY RATE USING MOSAICS WITH SIML

LINK: IOT-BASED SMART IRRIGATION MOTOR SYSTEM IEEE CONFERENCE PUBLICATION